On the scale bias of neural network training

Jaeho Lee @ POSTECH

LeT workshop
Oct. 7th, 2026

Disclaimer

Light on math, and light-hearted
(you'll have enough math today & tomorrow)

Motivation

“Efficient Learning Lab,” depuis 2022



Motivation

홈 > IT/과학

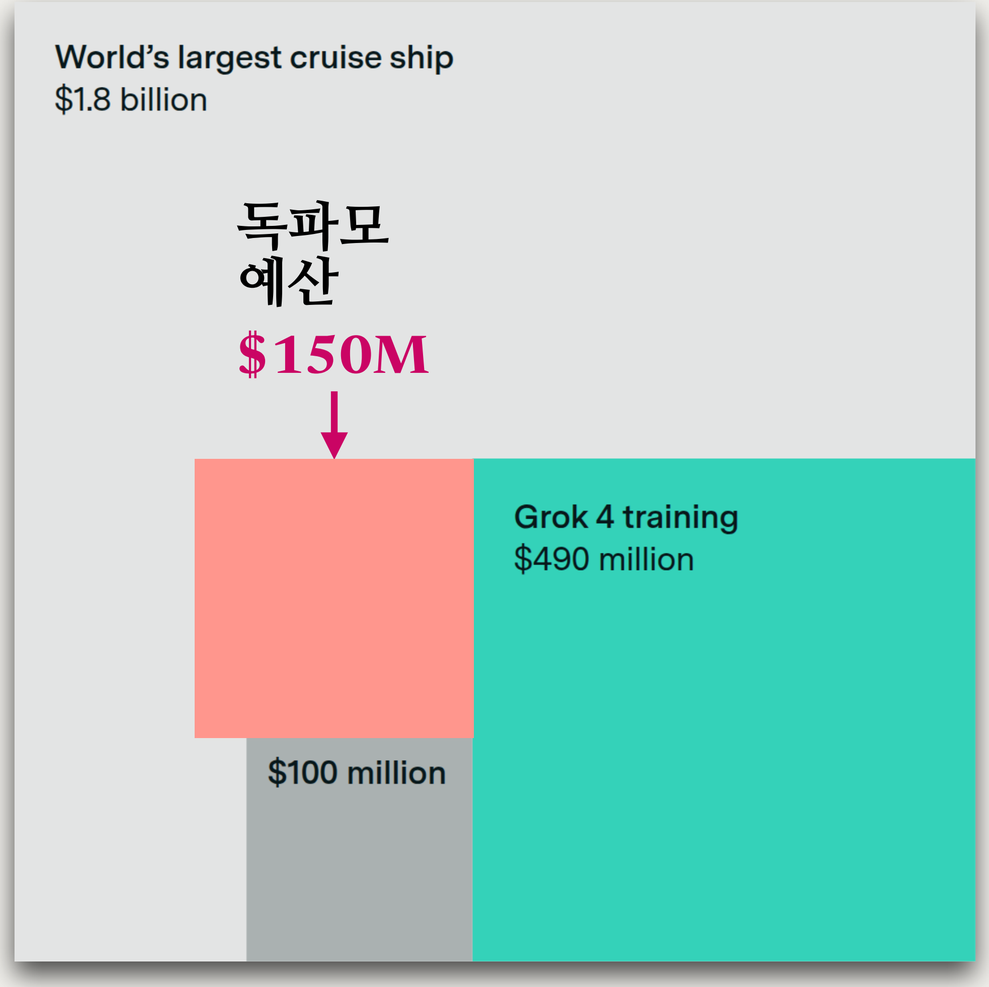
'국가대표 AI' 선발전 막 올랐다…이달 첫 탈락자 공개되나

윤지혜 기자 2026.01.05 15:55 구글에서 머니투데이 추가하기

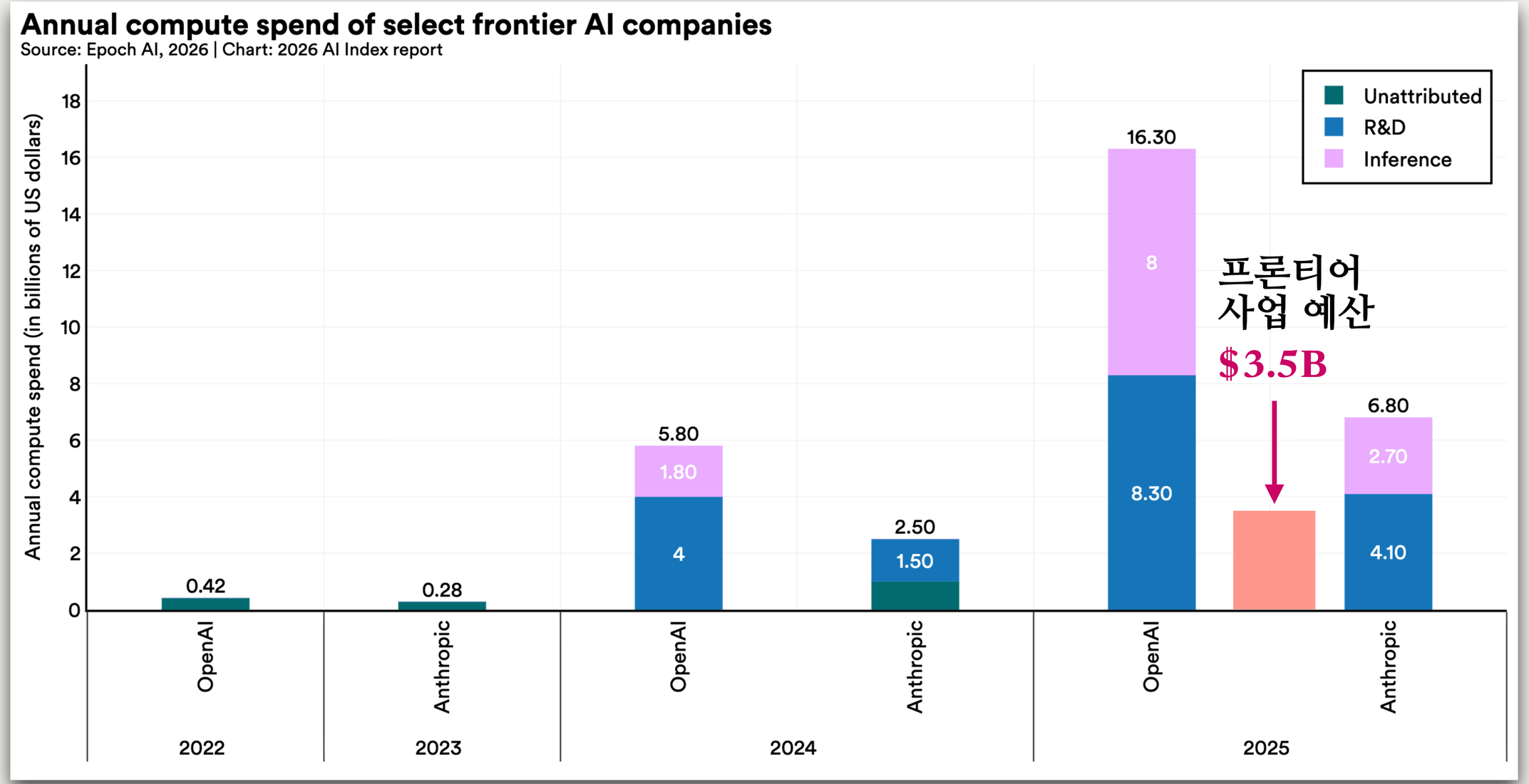
'15일까지 벤치마크 정량평가+전문가 정성평가 진행



Motivation

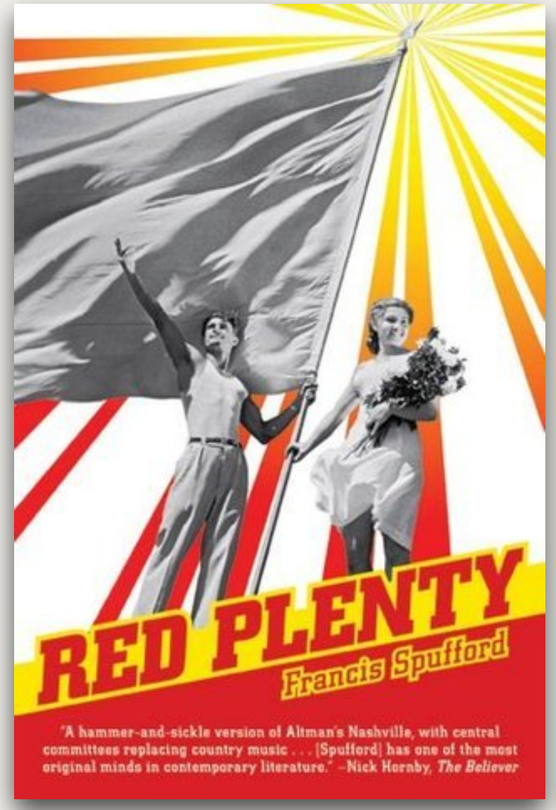

World's largest cruise ship
$1.8 billion
독파모
예산
$150M
Grok 4 training
$490 million
$100 million

Motivation


Annual compute spend of select frontier AI companies
Source: Epoch AI, 2026 | Chart: 2026 AI Index report
Annual compute spend (in billions of US dollars)
0
2
4
6
8
10
12
14
16
18
Unattributed
R&D
Inference
0.42
0.28
5.80
1.80
4
2.50
1.50
16.30
8
8.30
6.80
2.70
4.10
프론티어
사업 예산
$3.5B
OpenAI
Anthropic
OpenAI
Anthropic
OpenAI
Anthropic
2022
2023
2024
2025

Approach

- **Soviet spirit**
  - Who needs GPUs when you have math?







Approach

- Soviet spirit
  - Who needs GPUs when you have math?
- Who needs GPUs when you have brilliant people?
  - joint work with {태선, 상윤, 태혁}





Approach

Understanding ***implicit bias*** reveals opportunities for efficient training

Recap: Implicit bias

Image Credit: P. Nakkiran, “Deep Double Descent”, 2019

Implicit bias

- Learning is about **fitting functions to data**
  - **Example.** Fitting a degree-3 polynomial to 1D data

$$f(x) = a_3 x^3 + \cdots + a_0$$

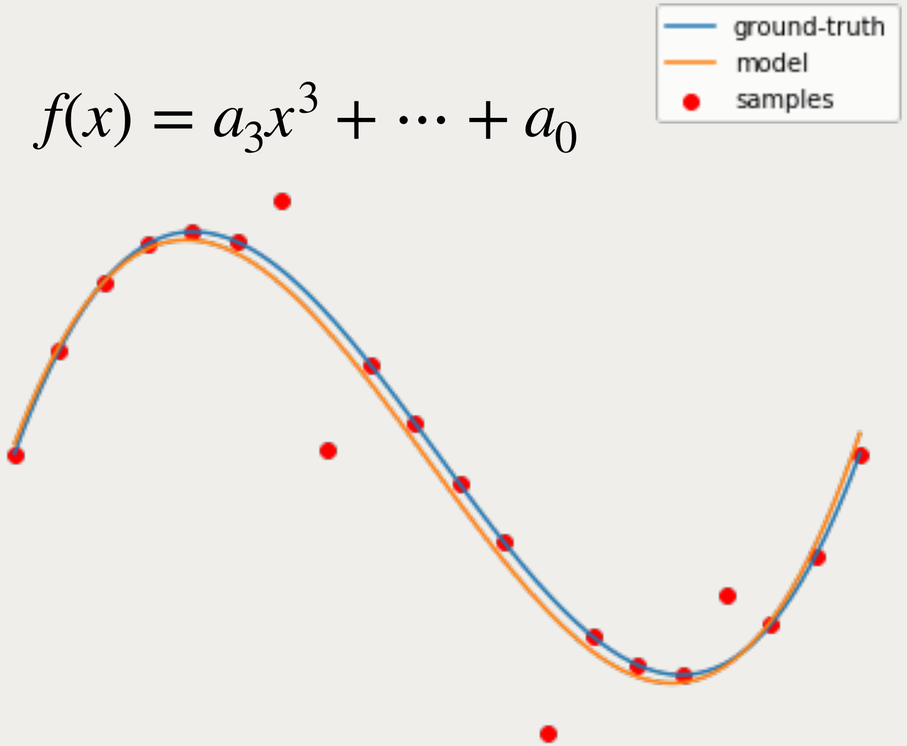


Image Credit: P. Nakkiran, “Deep Double Descent”, 2019

Implicit bias

- In deep learning, we typically have **a lot of parameters** – potentially trillions
  - imposing the risk of **poor generalization**

infinite possible solutions, as #params > #data

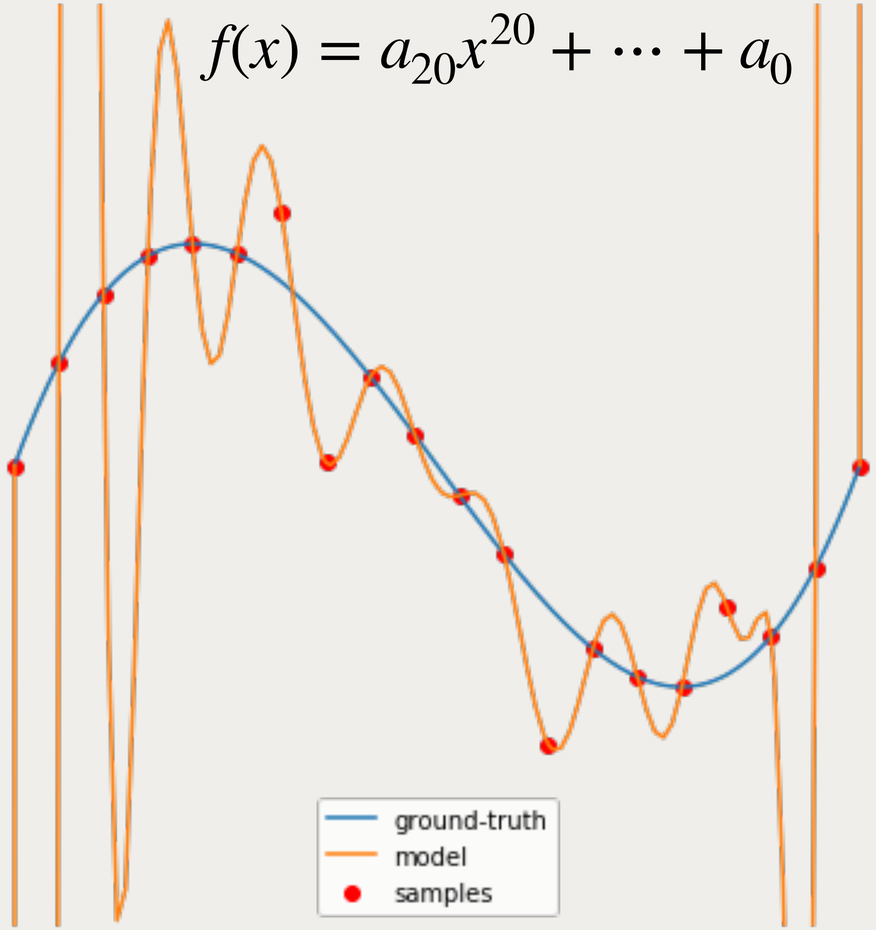
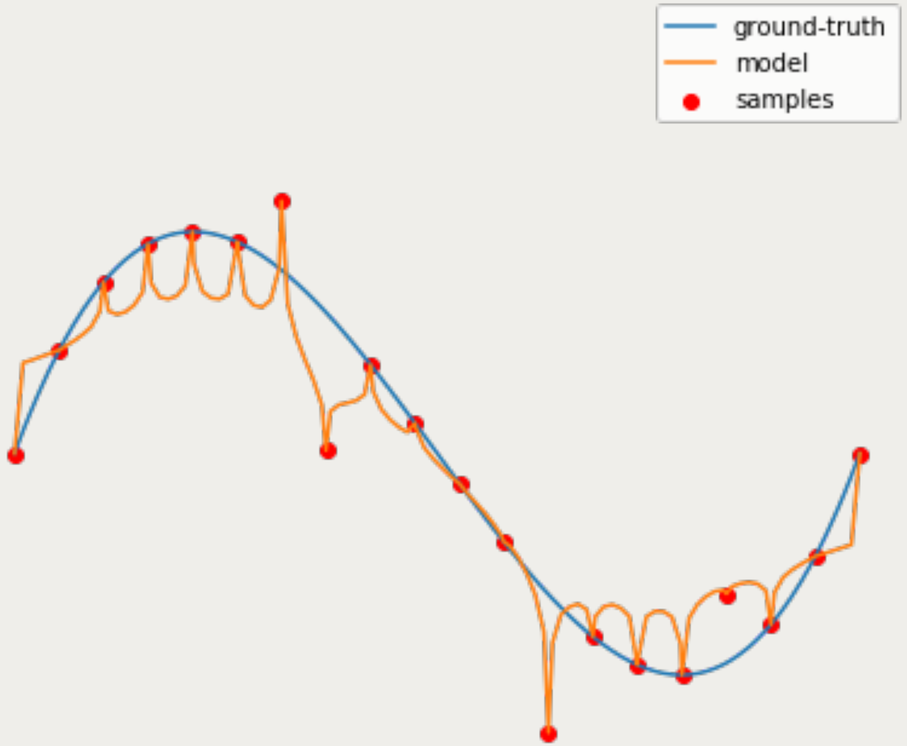


Image Credit: P. Nakkiran, “Deep Double Descent”, 2019

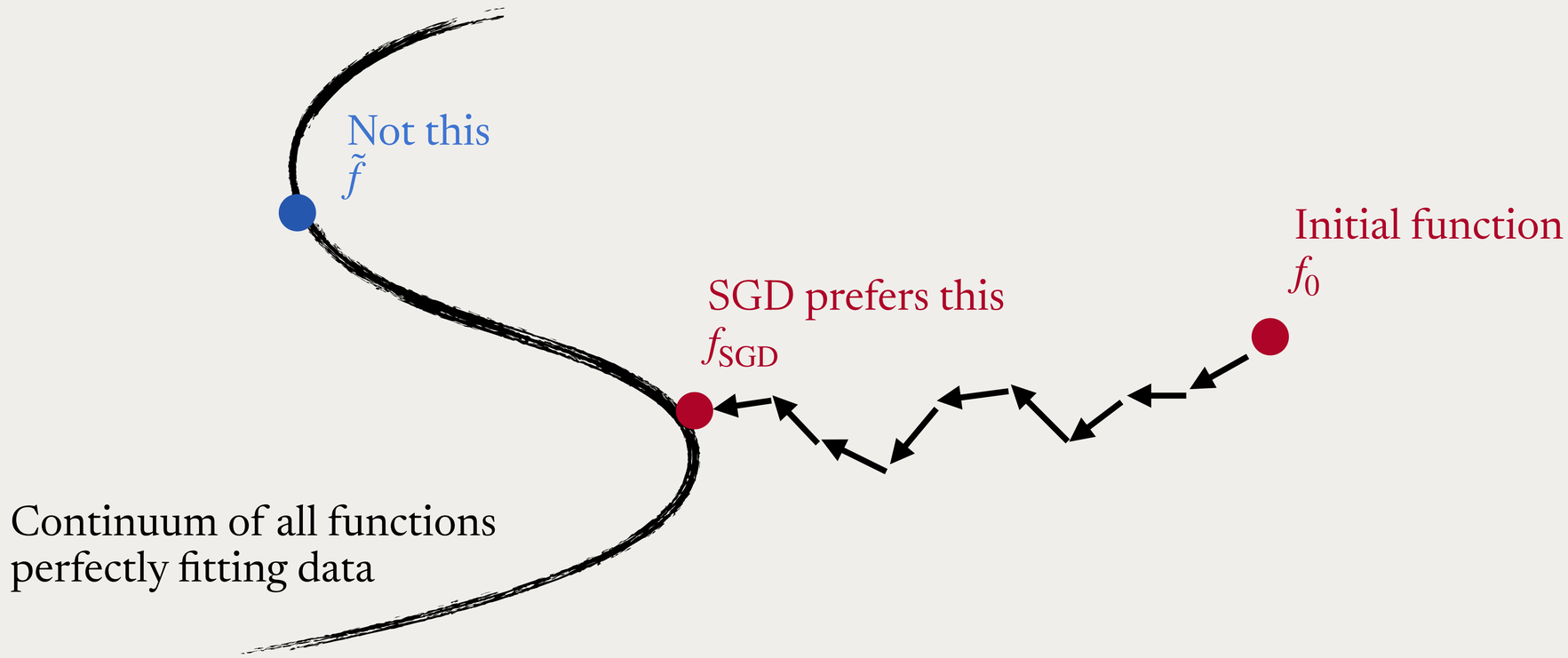
Implicit bias

- The **deep learning magic** is that neural nets automatically find solutions that generalize well
  - e.g., if we apply the **standard deep learning algorithms**...



Explanation

- SGD prefers learning **certain solution**
  - called **implicit bias**



Explanation

- The nature of this bias depends on (seemingly harmless) **hyperparameters**
  - Tuning HPs lead to different regularization effects (e.g., ℓ_1, ℓ_2)
    - Some lead to **good generalization**
    - Some lead to **bad generalization**

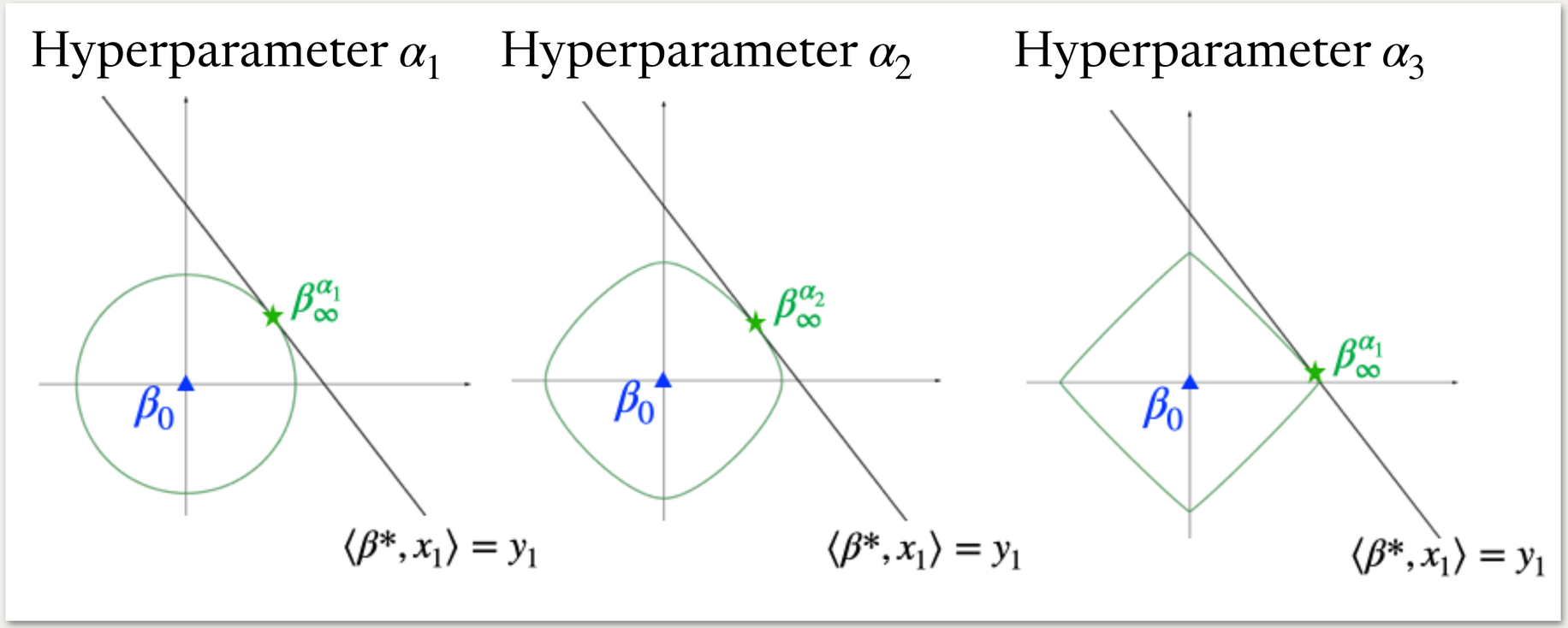
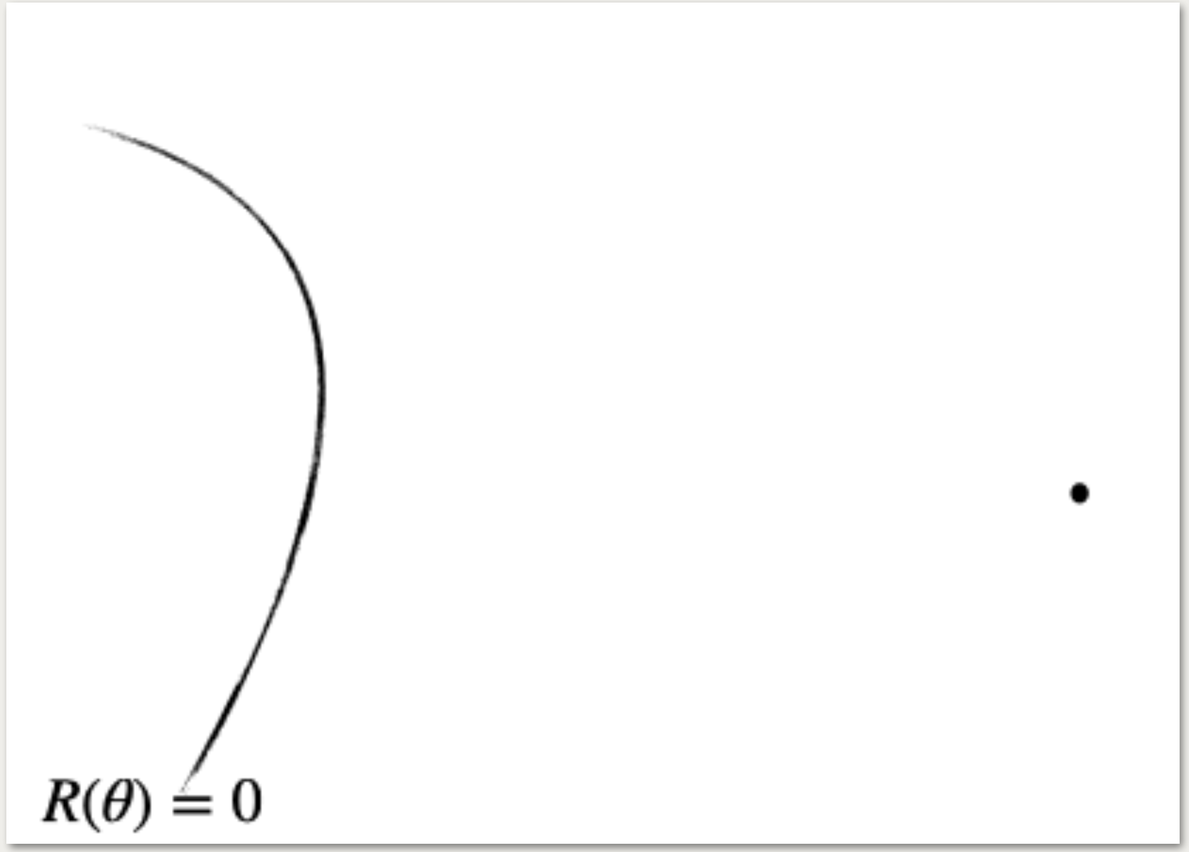


Image Credit: Pillaud-Vivien and Pesme, “Rethinking SGD’s noise – II: Implicit Bias”, 2022

Harnessing implicit bias

- **Rough intuition.** Implicit bias changes **how parameters travel** during training
  - Linked to the **training cost**?
  - Also changes **generalization**?



Scale bias

Question

How do we ***initialize*** a neural net, for the fast training?

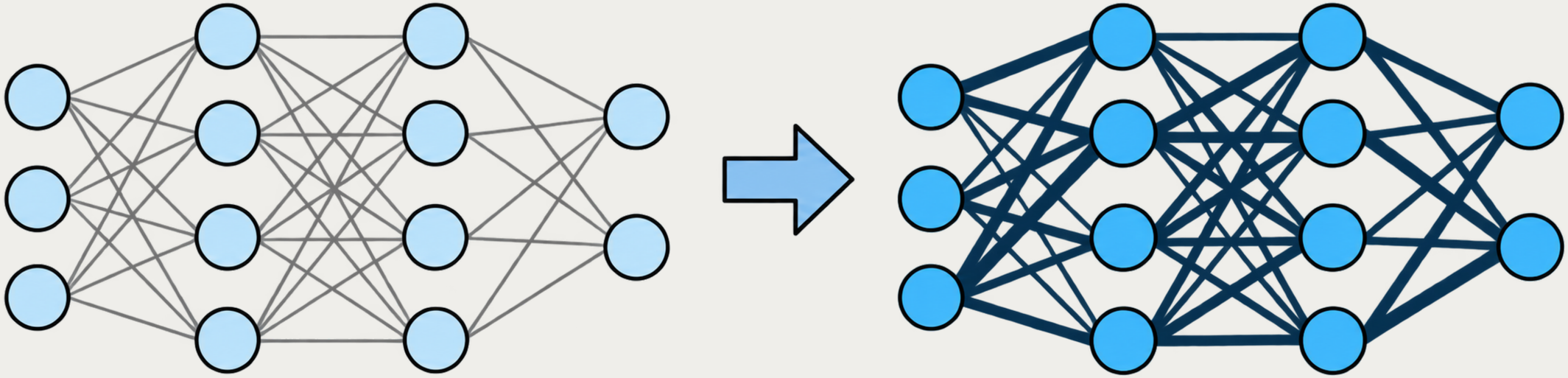
≈ tuning layerwise LR (or temperature)

Question

- Initial weights are typically drawn i.i.d. from the Gaussian

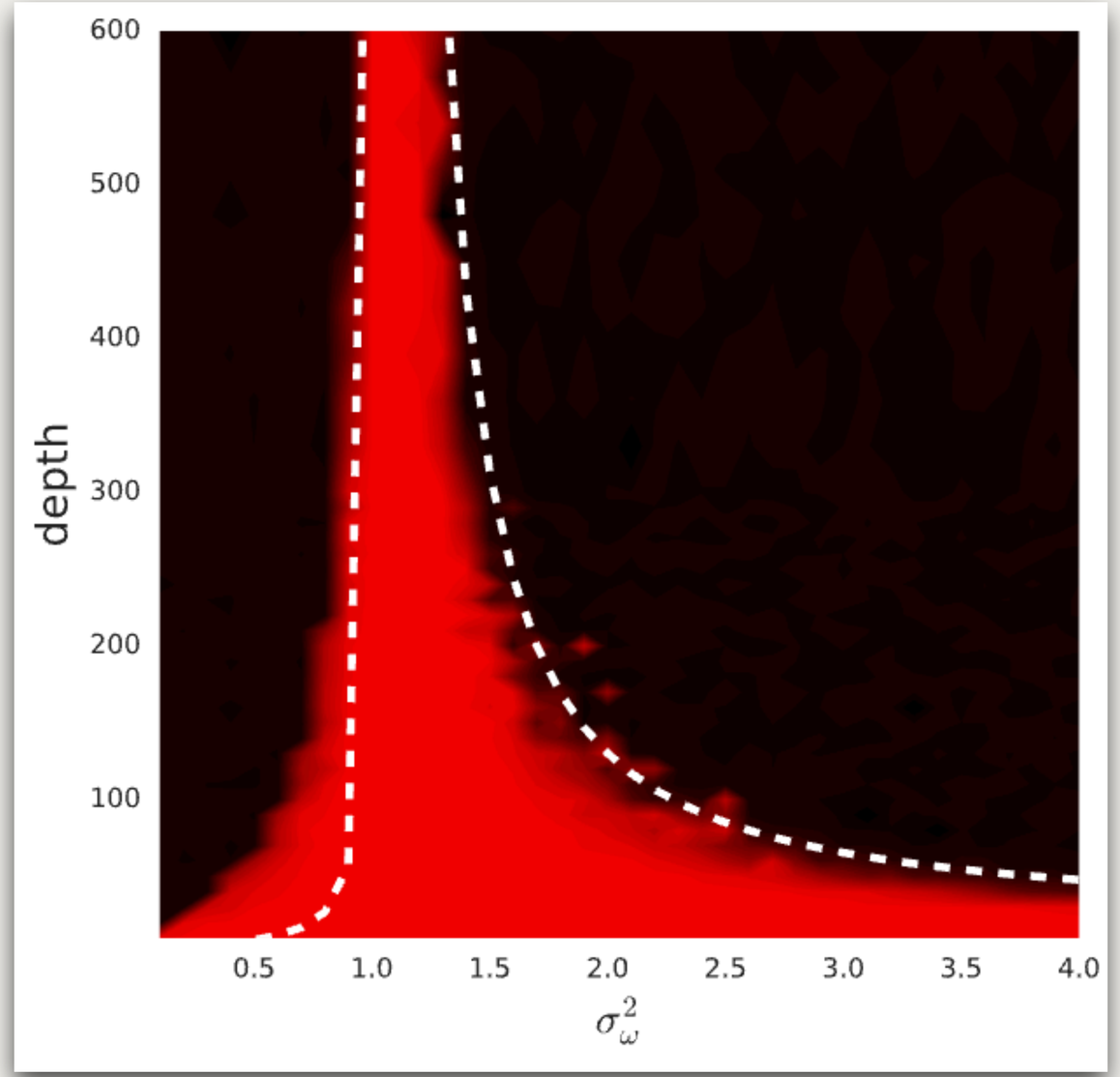
$$w_i \sim \mathcal{N}(0, \sigma^2)$$

  - Here, the **init. scale** σ^2 plays a key role – **scale bias**



Question

- Well-studied in **certain context**
  - Stabilize the training of ***very deep*** nets
    - Large σ^2: Activations explode
    - Small σ^2: Activations vanish
  - Studies **at-convergence** behaviors



Dynamical Isometry and a Mean Field Theory of CNNs:
How to Train 10,000-Layer Vanilla Convolutional Neural Networks

Lechao Xiao [1 2] Yasaman Bahri [1 2] Jascha Sohl-Dickstein [1] Samuel S. Schoenholz [1] Jeffrey Pennington [1]

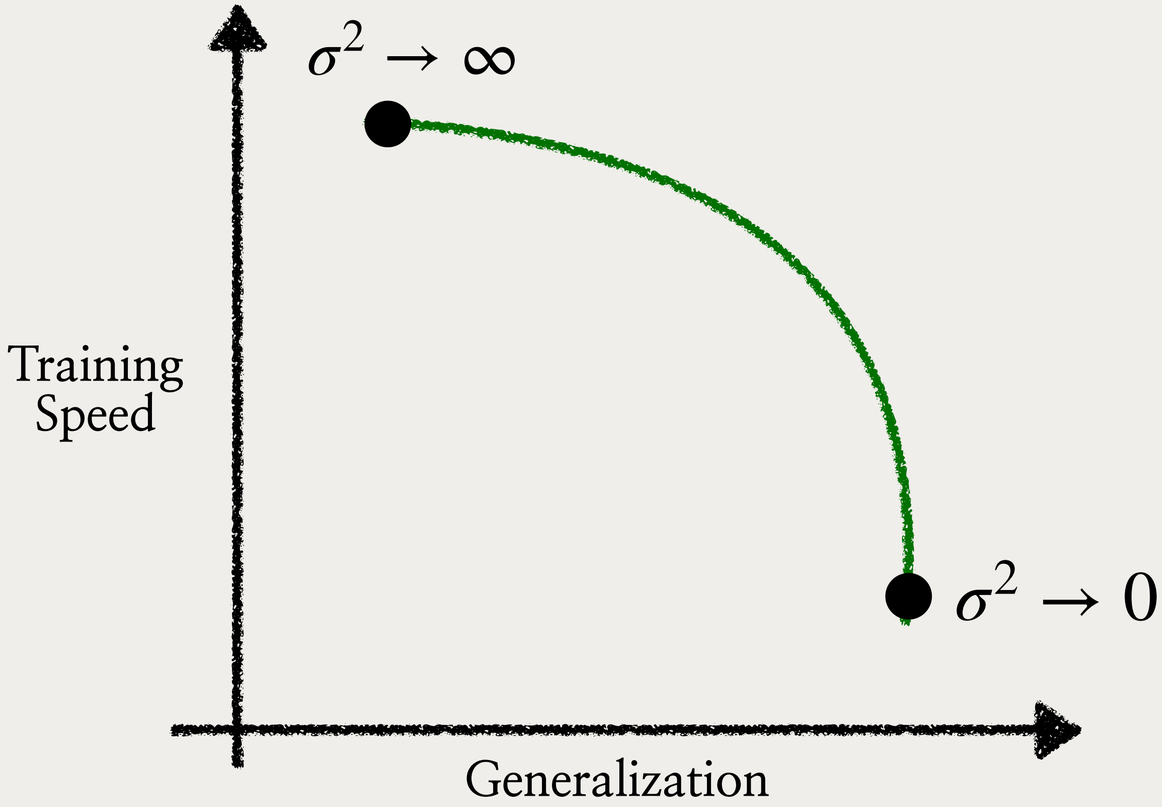
Question

How do we initialize a neural net, for the ***fast training***?

- Training dynamics, rather than at-convergence
- Not necessarily stacking 10,000 layers, w/o normalization layers

TL;DR

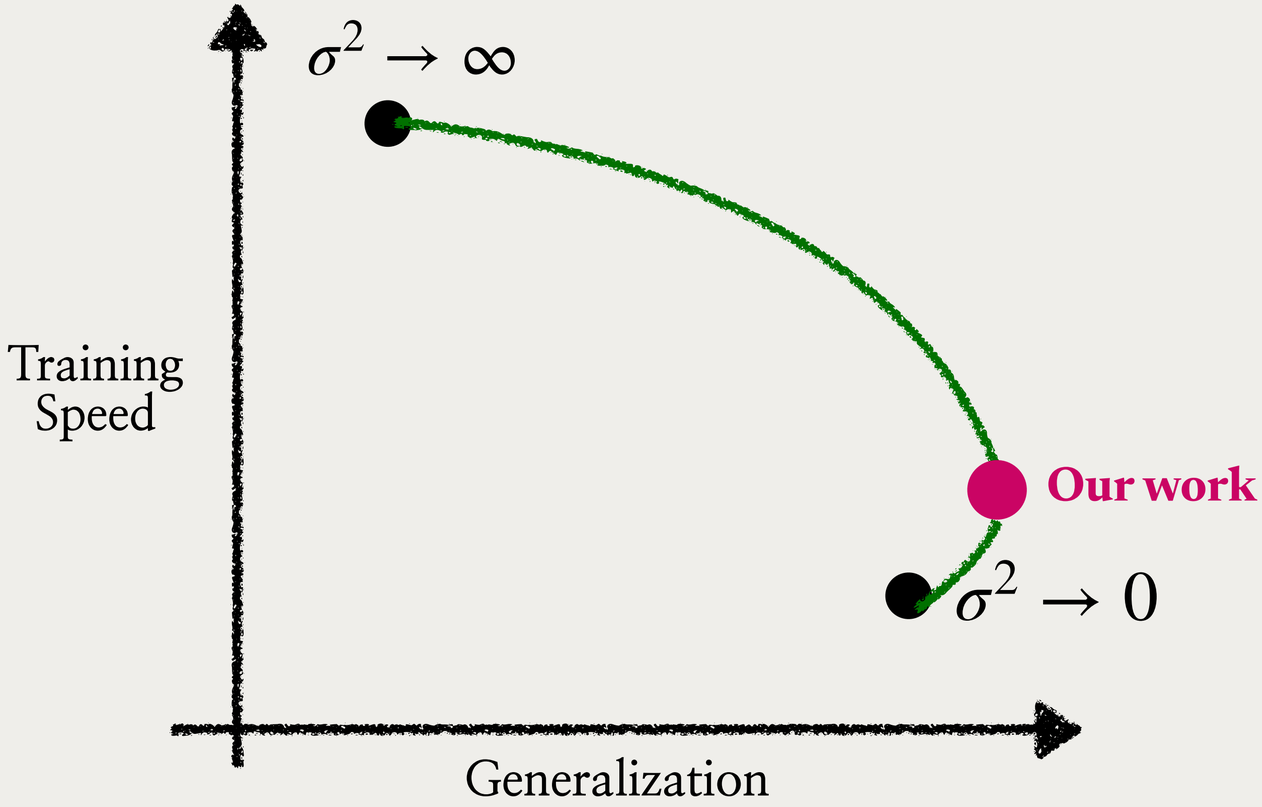
- Existing theories roughly capture the **speed-generalization tradeoff**
  - Thus, we thought we ***cannot harness this bias*** for free acceleration. But...



TL;DR

- We show that this is **wrong** – both empirically and theoretically
  - Exists a ***sweet spot*** where we can get the best of both worlds

up to 10x faster, for sinusoidal nets
up to 1.4x faster, for ConvNets



What was known

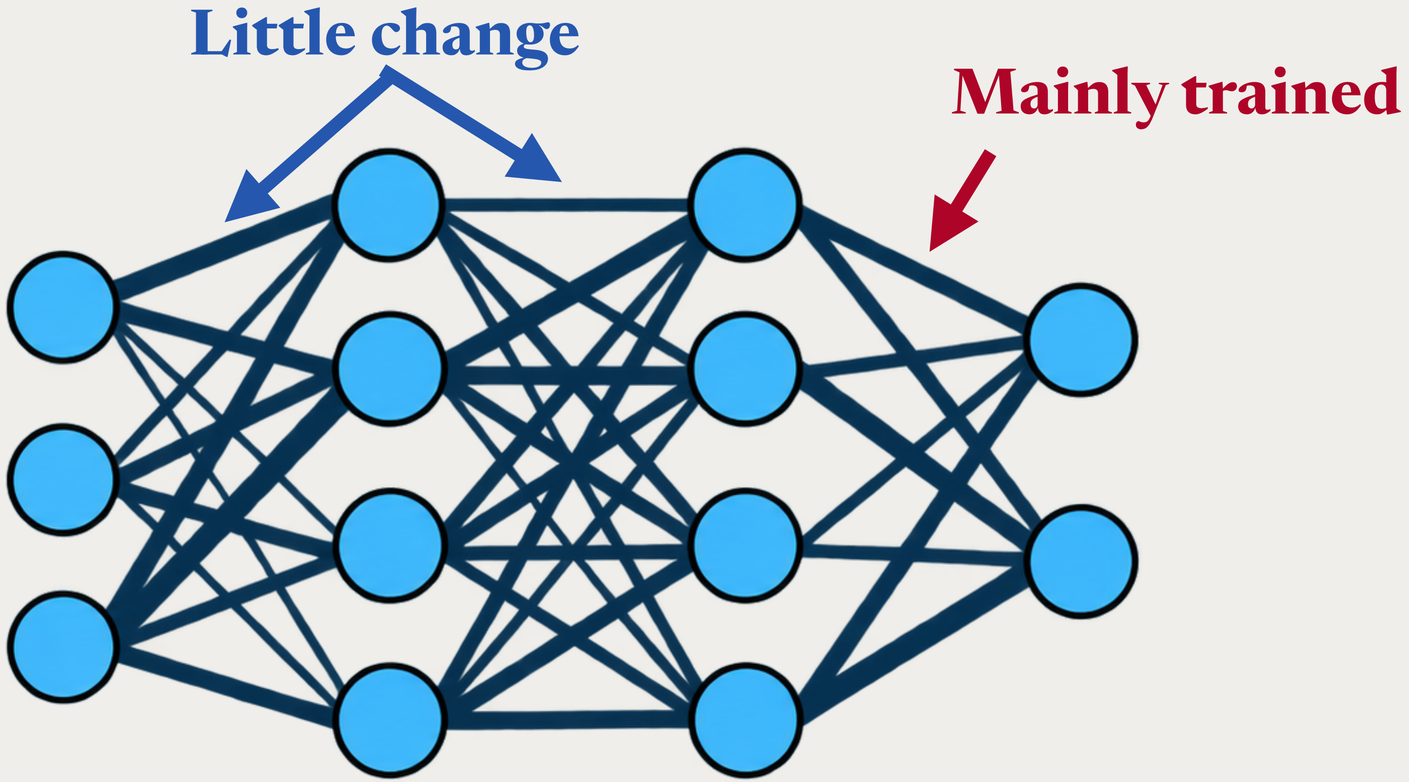
- Prior work study this at **two extremes**:
  - $\sigma^2 \to \infty$
  - $\sigma^2 \to 0$

- Popularly studied for two-layer ReLU nets

$$f(\mathbf{x}) = \sum_{i=1}^{m} a_i \cdot \mathrm{ReLU}(\mathbf{w}_i^\top \mathbf{x})$$

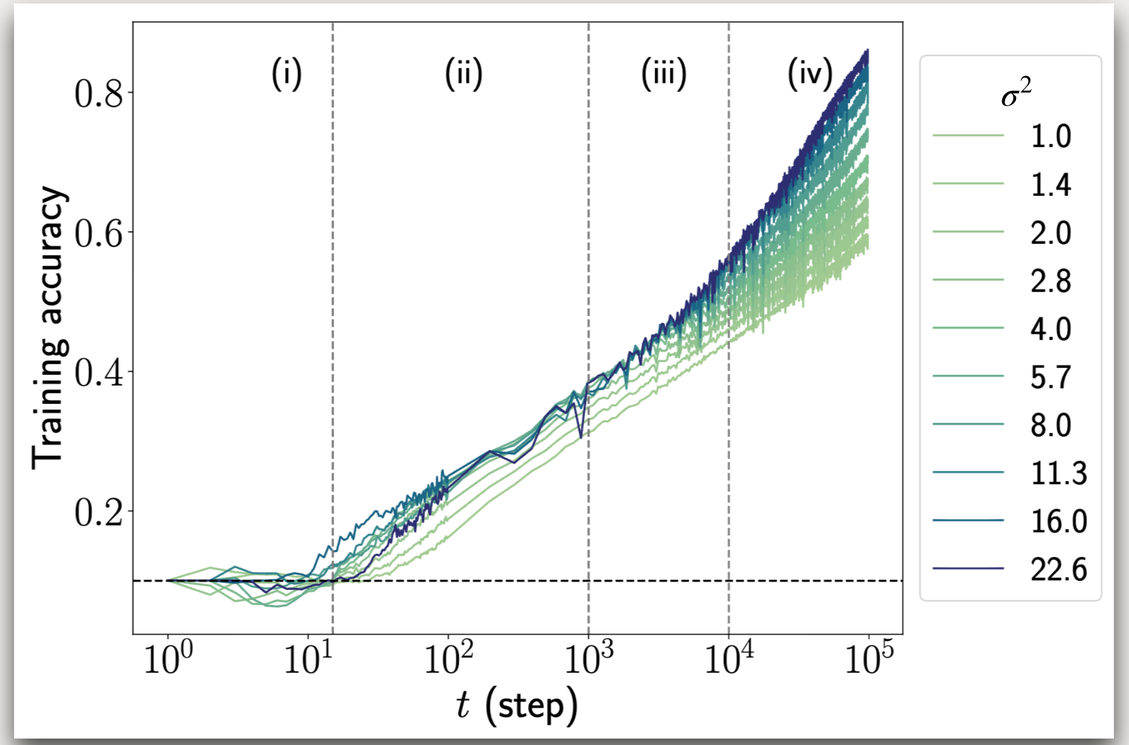
What was known

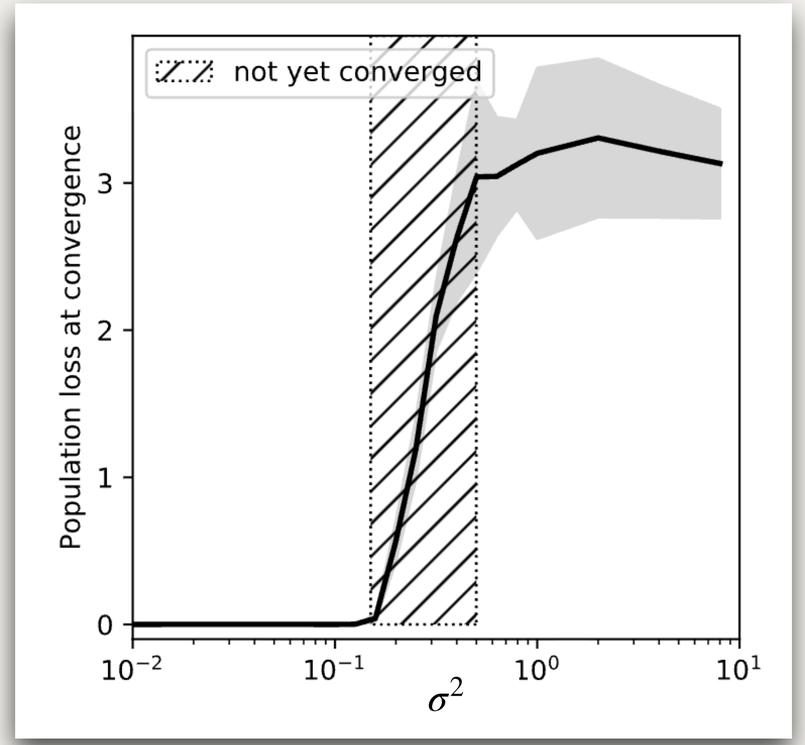
- **Very large σ^2.** Neural nets have **linearized dynamics** *a.k.a. kernel regime*



What was known

- **Very large σ^2.** Neural nets have **linearized dynamics**
  - **Fast optimization** – dynamics become much easier to analyze
  - **Poor generalization** – limited feature adaptation



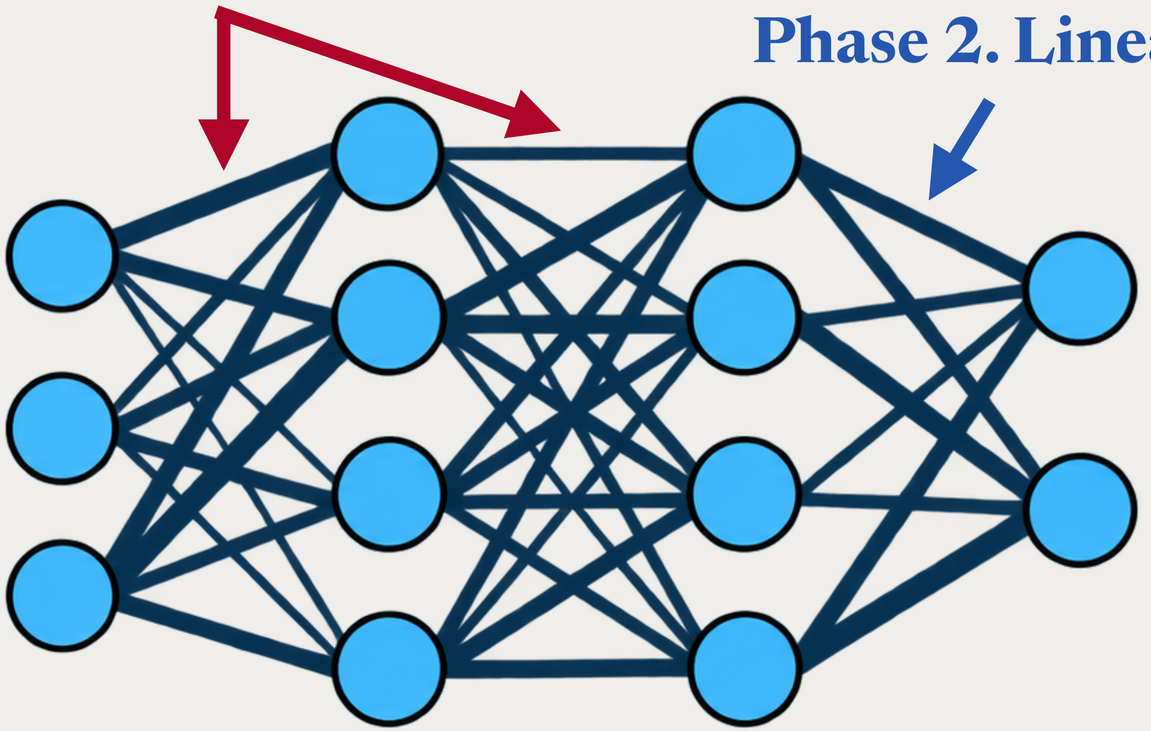


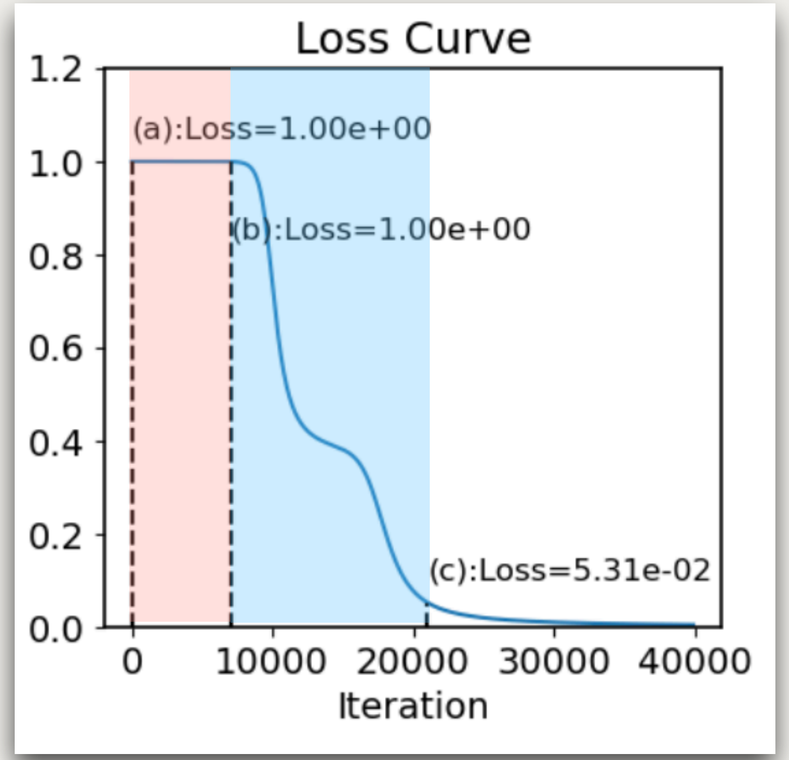
What was known

- **Very small σ^2.** Training proceeds in **two distinct phases**

Phase 1. Lower layer trains

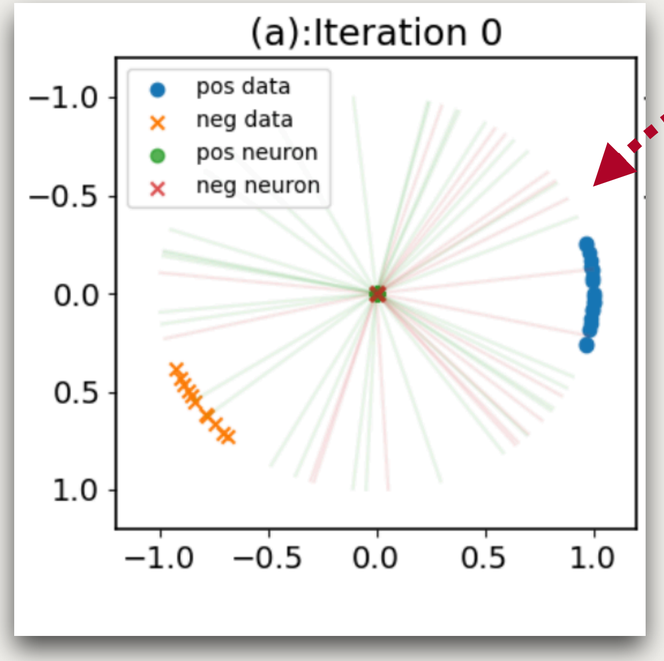
Phase 2. Linear model trains

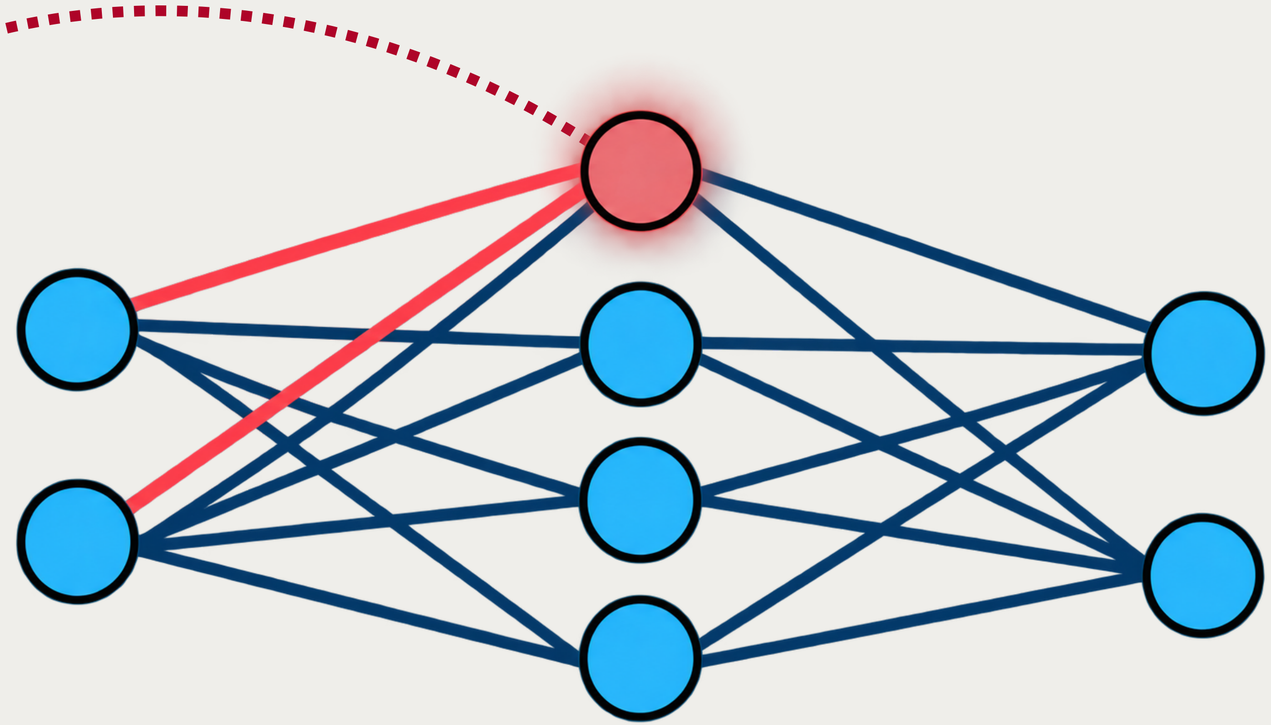




What was known

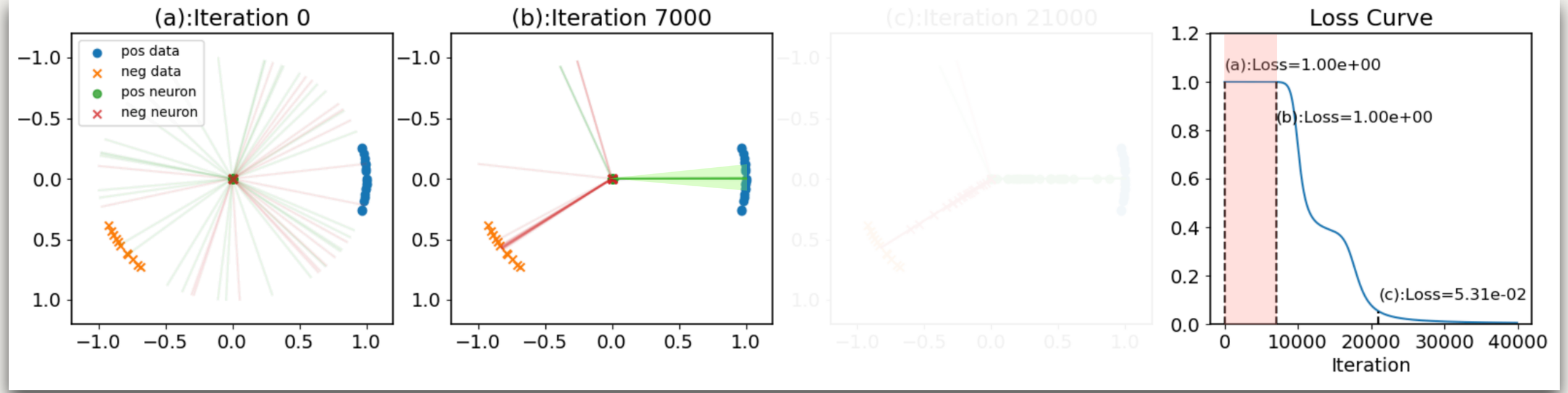
- **Very small σ^2.** Training proceeds in **two distinct phases**
  - **Initialization.** Each neuron can be represented by a vector – randomly spread





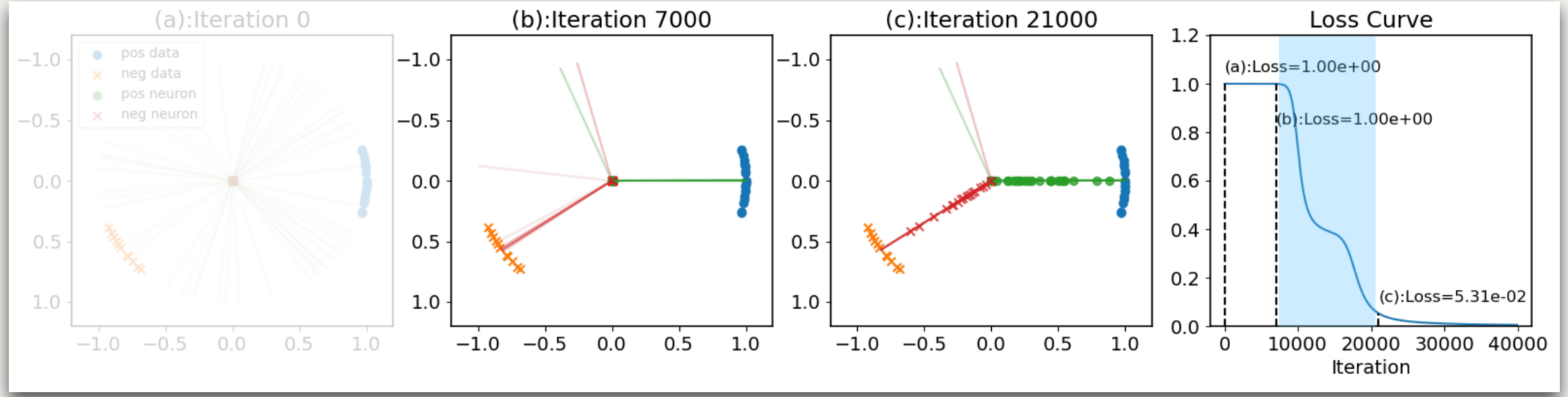
What was known

- **Very small σ^2.** Training proceeds in **two distinct phases**
  - **Initialization.** Each neuron can be represented by a vector – randomly spread
  - **Phase 1.** Neurons ***align*** to the data direction
    - Fits inside the **cone**, with the angle proportional to σ *requires $O(\log(1/\sigma^2))$ steps*



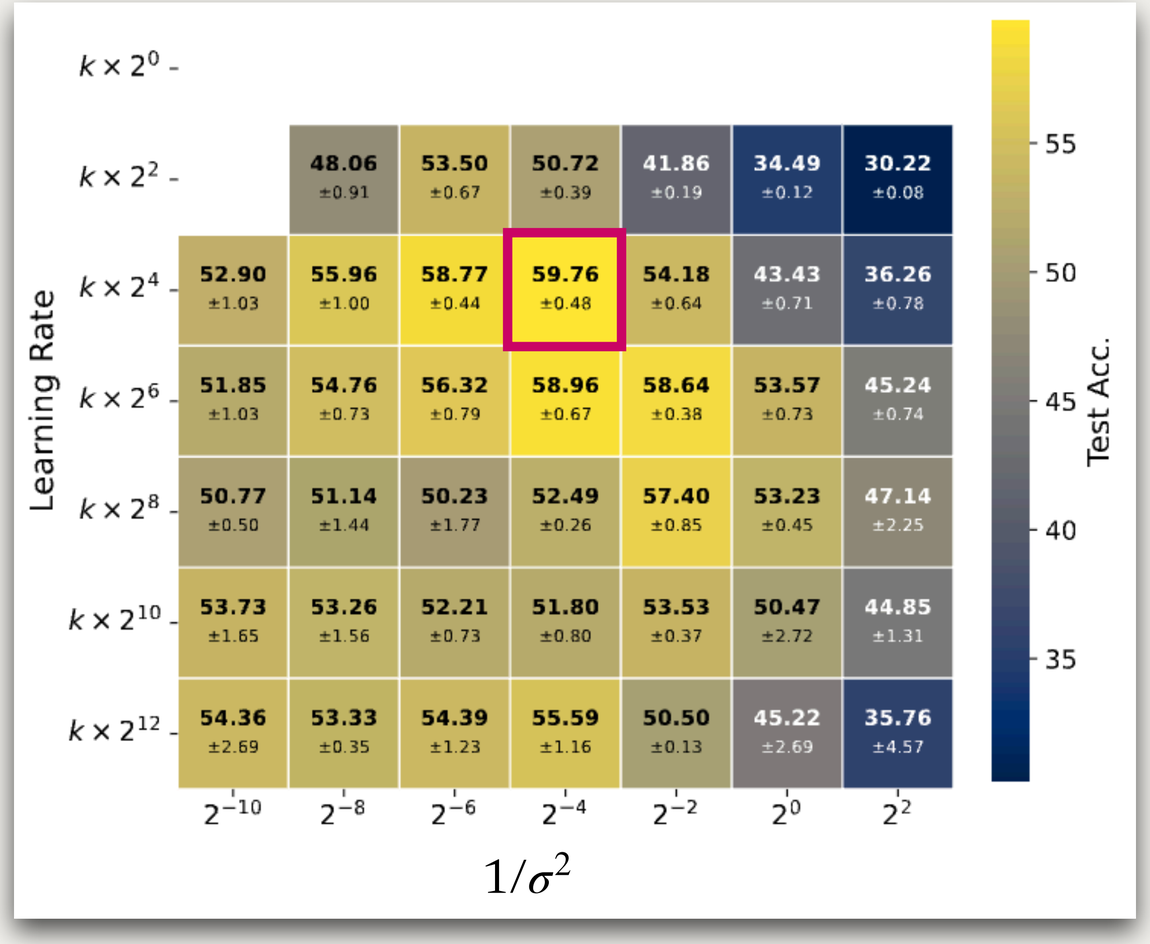
What was known

- **Very small σ^2.** Training proceeds in **two distinct phases**
  - **Initialization.** Each neuron can be represented by a vector – randomly spread
  - **Phase 1.** Neurons ***align*** to the data direction
  - **Phase 2.** Linearized dynamics



Empirical findings

- On practical neural nets, **smaller σ^2 ≠ better generalization**
  - The ***optimum*** is not in the extreme σ^2



Theoretical findings

- We provide the **first theoretical result** to explain this
  - Won’t go into details
  - Only some intuitions

Theorem 5.6. *Suppose that Assumption 4.1 holds,* $\|\hat{\mathbf{w}}_\alpha(t_{\eta,\alpha})\| \leq 1$, *and* $\langle \hat{\mathbf{w}}_\alpha(t_{\eta,\alpha}), \mathbf{x}_i \rangle \geq 0$ *for all* i *with* $y_i = +1$. *Let* G_ϵ *be the grid defined by* $G_\epsilon := \{-1 + k\epsilon : k \in \mathbb{Z}\} \cap [-1, 1]$. *Then, for any* $\delta \in (0,1)$ *and* $\epsilon \in (0, 0.25)$, *we have*

$$\mathcal{E}(\hat{\mathbf{w}}_\alpha(t_{\eta,\alpha})) - \mathcal{E}^* \leq \underbrace{\Phi\left(-\frac{\mathbf{v}_\star^\top \mathbf{s}_+}{\sigma}\right) - \Phi\left(-\frac{1}{\sigma}\right)}_{=\mathsf{OA}(\alpha)} \tag{26}$$

$$+ \underbrace{\frac{2(1+e)}{\sigma\sqrt{2\pi}}\left(g(\alpha)h(n,d) + \eta + C\left(1+\sigma\sqrt{d}\right)\sqrt{\frac{\log(6/\delta\epsilon)}{n}}\right)}_{\geq \mathsf{OF}(\alpha)}$$

with probability at least $1-\delta$, *where:*

$$\mathbf{v}_\star := \arg\max_{\mathbf{v} \in H(\alpha)} \mathbf{v}^\top \mathbf{s}_+, \tag{27}$$

$$g(\alpha) := 2\sqrt{2\pi} \cdot \sqrt{1 - r(\alpha)^2}, \tag{28}$$

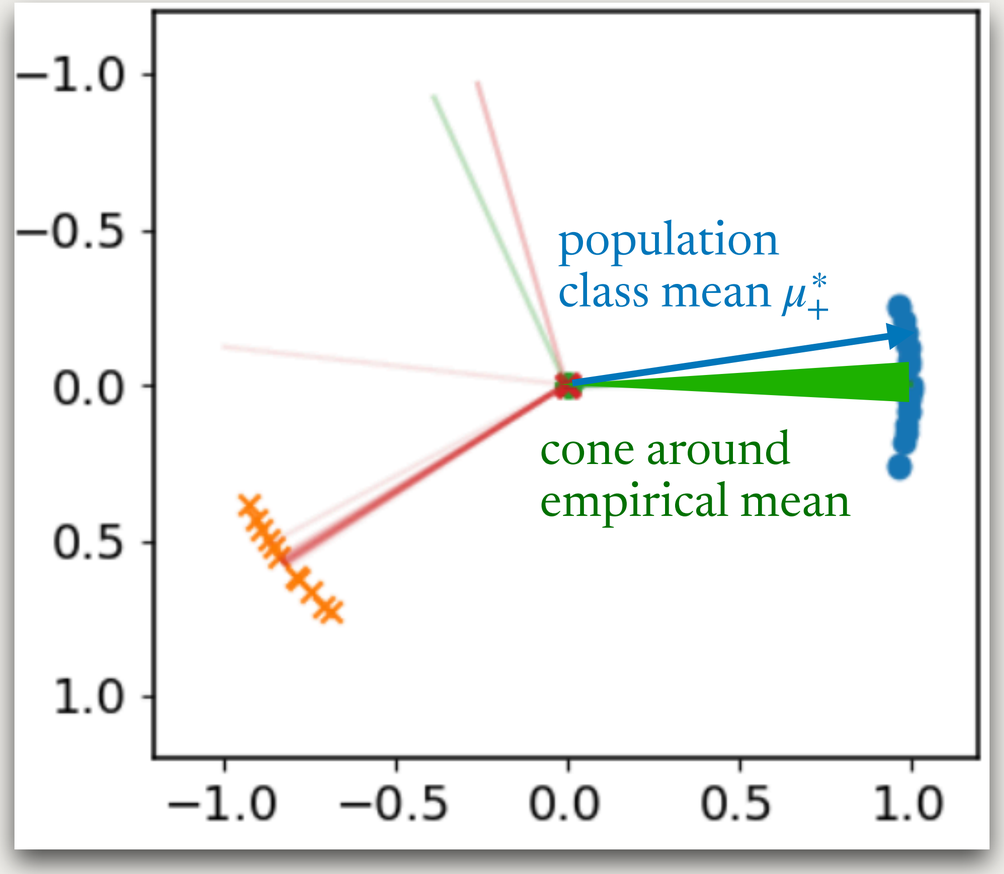
$$h(n,d) := \sqrt{\frac{d}{n}}\left(\sigma\sqrt{\frac{d}{n}} + \sigma + 1\right), \tag{29}$$

$$r(\alpha) := \max\left\{r \in G_\epsilon : 0 < r \leq \min_{\mathbf{v} \in H(\alpha)} \mathbf{v}^\top \mathbf{s}_+\right\} \tag{30}$$

and for some constants $C > 0$.

Intuition

- **Over-alignment** can happen in phase 1
  - Small $\sigma^2 \rightarrow$ **too narrow cone** $\rightarrow$ may not contain **population mean**, when samples are too few



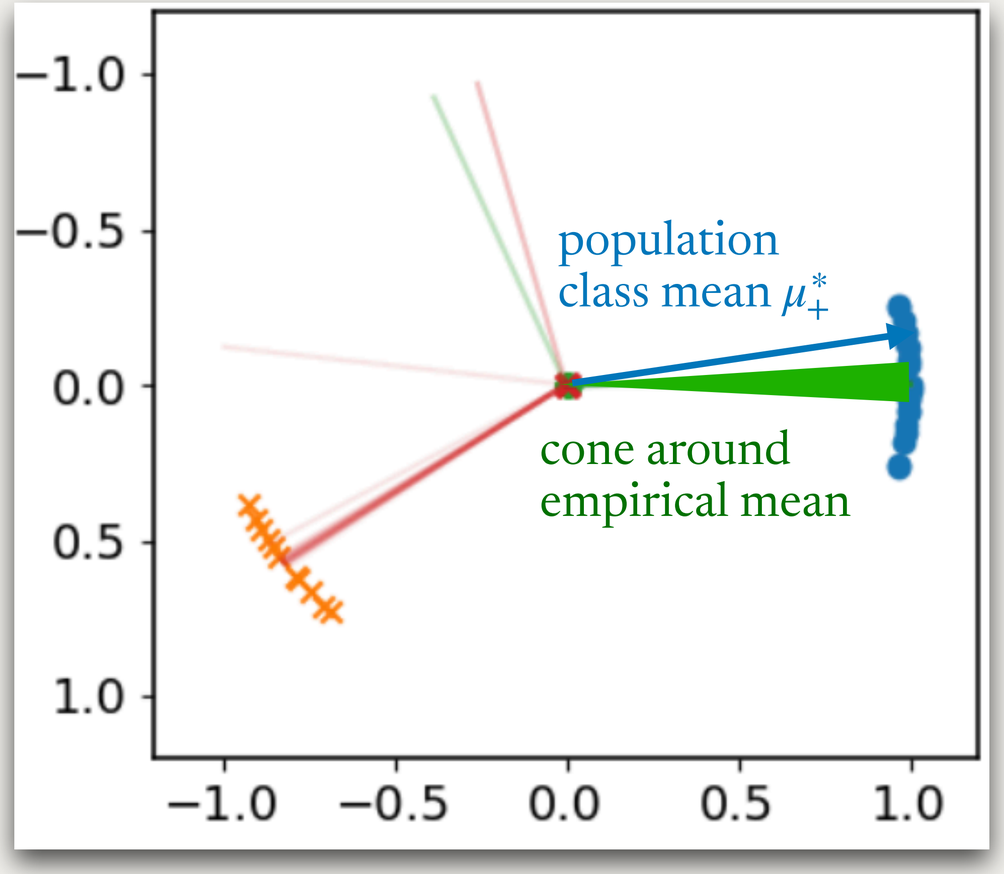
Intuition

- **Over-alignment** can happen in phase 1
  - Small $\sigma^2 \rightarrow$ **too narrow cone**
    - may not contain **population mean**, when n is small

- Excess error is:

$$R(\hat{\mathbf{w}}) - R(\mathbf{w}^*) = \inf_{\mathbf{w} \in \text{cone}} R(\mathbf{w}) - R(\mathbf{w}^*) + R(\hat{\mathbf{w}}) - \inf_{\mathbf{w} \in \text{cone}} R(\mathbf{w})$$

  - **Red:** “Over-alignment”
    - Larger for smaller σ
  - **Blue:** “Over-fitting”
    - Smaller for smaller σ



Intuition

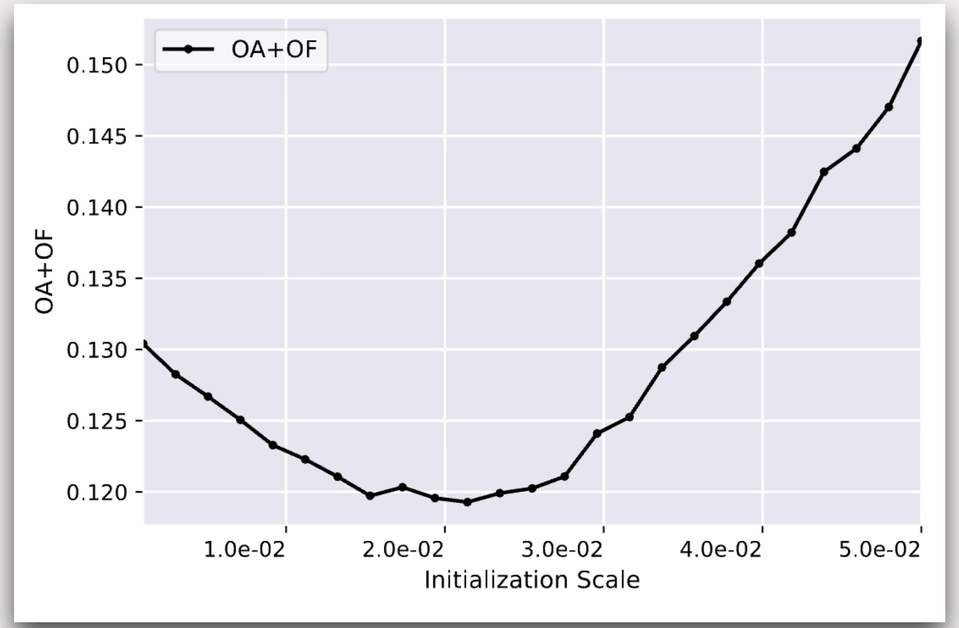
- Quantitative analysis gives us this theorem:

Theorem (informal). For sufficiently small σ, we have

$$\text{population risk} \leq \mathsf{OA}(\sigma) + \mathsf{OF}(\sigma)$$

  - $\mathsf{OA}(\sigma)$ decreases with σ
  - $\mathsf{OF}(\sigma)$ increases with σ

- Gives a nice U-shaped bound on the population risk
  - and thus we have the **sweet spot**



Wrapping up

- People thought **scale bias cannot be used** for faster training, unless we sacrifice accuracy
- We show that, **we can accelerate & generalize**

- **To-do.** Find the training-free recipe for finding optimal σ^2
  - Extend it to LLM-scale models

- More details at: https://arxiv.org/abs/2602.00827 (ICML 2026)

End-of-Slides

Takeaways

- Hyperparameter choices affects both training speed & generalization
- Theoretical analyses on the implicit bias may reveal some chances to improve the training speed, while preserving (or improving) generalization
- Thus we should spend more time on theory!

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